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Behavioral/Systems/Cognitive

Gating of Signal Propagation in Spiking Neural Networks by Balanced and Correlated Excitation and Inhibition

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Both ongoing and natural stimulus driven neuronal activity are dominated by transients. Selective gating of these transients is mandatory for proper brain function and may, in fact, form the basis of millisecond-fast decision making and action selection. Here we propose that neuronal networks may exploit timing differences between correlated excitation and inhibition (temporal gating) to control the propagation of spiking activity transients. When combined with excitation–inhibition balance, temporal gating constitutes a powerful mechanism to control the propagation of mixtures of transient and tonic neural activity components.

Introduction

Appropriate action selection is one of the fundamental problems that the brain needs to solve in its interaction with the environment. This problem is even enhanced by the intrinsically parallel architecture of the brain. A mechanism based on winner-take-all dynamics is often suggested for this purpose (Ermentrout, 1992). However, winner-take-all dynamics require strong recurrent inhibition, which may not be available in every brain region (Tepper et al., 2004). Moreover, winner-take-all dynamics are typically slow if the competing signals are of comparable strength, and may even result in unpredictable responses in the presence of transient synchrony within and among the competing inputs (Ermentrout, 1992). Another approach to action selection could be based on the gating of signals, such that only one (or few) effectively propagate to the next processing stage. Thus, recently Vogels and Abbott (2009) demonstrated that detailed balance of excitation and inhibition (amplitude gating) can efficiently gate the propagation of firing rates. However, while this mechanism allows for a rapid selection among competing rate inputs, it fails to control activity transients on a timescale of up to a few tens of milliseconds [Vogels and Abbott (2009), their Fig. 3*f*]. In a natural stimulus environment, such activity transients frequently occur as a consequence of stimulus-driven changes in firing rates or by the emergence of correlations in the network (Buracas et al., 1998; Bar-Yosef and Nelken, 2007; Butts et al., 2007; Hromádka et al., 2008; Haider et al., 2010). Therefore, reliable gating of activity transients is a prerequisite of proper brain function.

Here we propose an alternative gating mechanism that exploits temporal differences between correlated excitation and inhibition (Okun and Lampl, 2008; Gentet et al., 2010). We show that small differences between the timing of excitation and inhibition (referred to temporal gating from hereon) can act as a modulator of neuronal integration time and, thereby, serve as a highly selective gate for signal transients. The proposed mechanism allows for both gating at a millisecond timescale and reliable control of onset transients and correlated rate inputs. When combined with amplitude gating, temporal gating constitutes a powerful mechanism to control the propagation of mixtures of transient and tonic neural activity components.

Materials and Methods

Models

Neurons

The neurons in the network were modeled as leaky integrate-and-fire neurons, with the subthreshold dynamics of the membrane potential $V^i(t)$ in neuron i described by the following equation:

$$C\frac{d}{dt}V^i(t) + G_{\text{rest}}[V^i(t) - V_{\text{rest}}] = I^i_{syn},$$

where I^i_{syn} is the total synaptic input current into neuron i, and C and G_{rest} denote the passive electrical properties of its membrane at rest (V_{rest}). When the membrane potential reached a fixed spike threshold V_{th} above rest, a spike was emitted, the membrane potential was reset to its resting value, and synaptic integration was halted for 2 ms, mimicking the refractory period in real neurons. The parameters used in the simulations were as follows: $C = 290$ pF, $G_{\text{rest}} = 29$ nS, $V_{\text{rest}} = -70$ mV, and $V_{\text{th}} = -57$ mV (Destexhe et al., 1998; Muller et al., 2007).

Network

We simulated a network of 28,125 neurons, with 22,500 excitatory and 5625 inhibitory neurons (ratio 4:1). The excitatory neurons were arranged on a Cartesian grid of 150×150, the inhibitory neurons on a grid of 75×75. Both grids represented the same cortical space of 1×1 mm^2. To avoid boundary effects, the network was folded as a torus. The recurrent network was sparsely connected with a connection probability of ~5% ($K_{\text{exc}\to\text{exc}} = 1120$, $K_{\text{exc}\to\text{inh}} = 280$, $K_{\text{inh}\to\text{exc}} = 1120$, $K_{\text{inh}\to\text{inh}} = 280$). As an exception to this rule, excitatory neurons that participated in the signal path received less than 5% connection from the background neurons, to maintain the same in-degree. The local connection probabil-

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E01 瞬变门控被定为前提问题
作者报告｜作者把瞬变的选择性门控当作前提而非可选项，为整篇文章的必要性论证定调，也说明为什么只用时间平均放电评价门控是不够的。

E09 毫秒级决策的动机
作者报告｜把机制与动作选择场景挂钩，是全文的动机来源，但也是最需要在引用时与计算证据分开的一层。

E02 核心主张：利用 E/I 时序差异
作者报告｜这是本文机制的一句话表述：门控的旋钮是时序而不是权重或平衡本身。

E07 缺口陈述的完整形式
作者报告｜摘要给出的是合用主张而非单独主张，这一点决定了时间门控的定位：它是补充轴，不是替代方案。

E03 对 winner-take-all 的第一条批评
作者报告｜说明本文为什么不去走竞争抑制这条更常见的门控路线，为选择时序门控提供动机上的合理性。

E04 winner-take-all 在强度接近时很慢
作者报告｜作者用来支持毫秒级门控需求的第二条理由，与本文反复强调的瞬时性直接相关。

E11 神经元模型类型
作者报告｜点神经元 LIF 是全部分析的基础，也决定了 τ_eff 的含义是膜电位积分的时间尺度。

Ｅ０５　既有机制：幅度门控
作者报告｜明确指出前作机制的有效范围，即发放率门控是有效的，为后面的对照做准备。

Ｅ０６　知识缺口：幅度门控关不住瞬变
作者报告｜本文的核心缺口陈述，同时给出了失效的时间尺度量级（几十毫秒），引用时应保留这个量级。

Ｅ１２　不应期设置
作者报告｜２　ｍｓ　不应期与本文关心的几毫秒时间尺度同量级，是评价同步活动能否越过分界线时必须纳入的因素。

Ｅ０８　替代机制的提出
作者报告｜把时序门控定位为　ａｌｔｅｒｎａｔｉｖｅ，说明它与　ｄｅｔａｉｌｅｄ　ｂａｌａｎｃｅ　并列而非升级。

Ｅ１３　网络规模与　Ｅ／Ｉ　比例
作者报告｜给出大规模随机递归网络的规模与　４：１　比例，说明结果不是在极小网络中得到的。

Ｅ１４　通路神经元的连接补偿
作者报告｜通路神经元为保持入度而减少背景输入，说明作者控制了入度这一混杂变量，但也意味着通路神经元的工作点被人为拉齐。

ity was described as a function of their cell body distance using a Gaussian profile (Hellwig, 2000). The width of the Gaussian describing the connection probability σ_e was set to 0.6 mm for the excitatory and $\sigma_i = 0.1$ mm for the inhibitory connections (Stepanyants et al., 2008) (Fig. 1*a*). In addition to the local connections from within the network, each neuron received excitatory connections from outside, mimicking both long-range connections within the same cortical area and input from other areas in cortex (Braitenberg and Schüz, 1991; Kumar et al., 2008a). The external inputs were modeled as uncorrelated Poisson-type spike trains. The rate and strength of external inputs (1500 excitatory synapses per neuron, each at ~2 Hz) were adjusted to obtain a low-firing-rate (~3 Hz), asynchronous (pairwise correlation ≈ 0.01), and irregular activity state in the network (Brunel, 2000; Kumar et al., 2008b).

Synapses

Synaptic inputs were modeled as transient conductance changes, using exponential functions with $\tau_{exc} = 1.5$ ms and $\tau_{inh} = 10$ ms (Kuhn et al., 2004; Muller et al., 2007). Excitatory and inhibitory synaptic delays in the embedding network were set to 2 ms. The weights (peak conductance) of the excitatory synapses targeting excitatory neurons (J_{EE}) were set to 0.5 nS, such that a postsynaptic potential of ~0.15 mV was elicited at rest (Matsumura et al., 1996; Bruno and Sakmann, 2006; Kumar et al., 2008a). The weight of excitatory synapses targeting inhibitory neurons (J_{EI}) was set to 1 nS (Cruikshank et al., 2007; Kremkow et al., 2010). The synaptic weights of all inhibitory synapses (J_{II}, J_{IE}) were set to a 0.5 nS conductance value. This set of synaptic strengths and external input (see above) generated both a realistic effective membrane time constant of ~5 ms (Anderson et al., 2000; Destexhe et al., 2003; Léger et al., 2005) and a realistic low-firing-rate, asynchronous-irregular network activity regime (Brunel, 2000; Kumar et al., 2008b; Ecker et al., 2010).

Embedding of a feedforward network as signal path

To construct a signal path, we embedded a feedforward network, containing three successive groups of neurons (sender, gate, and receiver groups), into the recurrent network. The sender group was composed solely of 100 excitatory neurons, whereas the gate and receiver group contained 100 excitatory and 25 inhibitory neurons each (Fig. 1*b*). The inhibitory neurons provided the balancing inhibition in the signal path and controlled the effective integration time of the excitatory neurons. The three groups were spatially separated to avoid cross talk between groups (Kumar et al., 2008a) (Fig. 1*b*). The neurons constituting a group were randomly chosen from a local pool of 300 excitatory and 75 inhibitory neurons. The center of the neuron pool defined the physical location of the group in the network. Each neuron in the signal path received convergent inputs from 60 randomly selected excitatory neurons of the preceding group. The neurons in the gate and receiver groups received excitatory synapses from 1060 neurons (<5%) of the background network to maintain uniform in-degree in the network. The delay of excitatory synapses in the signal path was set to 5 ms (Aertsen et al., 1996; Diesmann et al., 1999). This choice of the excitatory delays of 5 ms is somewhat arbitrary. These delays are supposed to mimic the latency between different neighboring brain regions, and such delays can be in the order of up to 10 ms. Further, we point out that these delays do not influence the dynamics of the embedding network and the embedded feedforward network, and can be reduced to any arbitrary value without influencing the gating. In the default state of the signal path, the delay of inhibitory synapses onto the excitatory neurons within a group was set to 2 ms (Cruikshank et al., 2007; Kremkow et al., 2010).

Embedding multiple signal paths

To construct two signal paths that merged in the gate group, we embedded the first signal path as described above. The sender group of the second signal path was then added. It was spatially separated from the first sender group to avoid cross talk. The excitatory neurons in the gate and receiver groups were common in both signal paths. The inhibitory neurons of the gate group in both signal paths were randomly chosen from a local pool of 75 inhibitory neurons. Thus, both signal paths might share a small fraction of inhibitory neurons in the gate group (see Fig. 5*a*).

Stimuli

We used three different stimulus classes to study signal propagation and gating in the signal path.

Pulse packet input. The input to the sender group was a volley of spikes with α spike times drawn from a Gaussian distribution of width σ (Aertsen et al., 1996; Diesmann et al., 1999). The number of synapses from the presynaptic neurons to the sender group was set to 60, the same as within the signal path.

Uncorrelated rate input. We used a Poisson process to stimulate the sender group with uncorrelated firing rate input (Vogels and Abbott, 2009). The onset time of the stimulus was set to 500 ms to mimic a sudden activation of the signal path (Butts et al., 2007). The rate of the Poison process was set to 200 Hz to induce a strong onset transient.

Correlated rate input. To introduce correlation in the firing rate input, we used the multiple-interaction process (MIP) (Kuhn et al., 2003). This process introduces correlations by copying spikes from a Poisson "mother process" into a desired number of "children" processes. The copy probability (c) defines the correlation among the children processes. As in the uncorrelated rate input, the onset time constant of the correlated rate stimulus was set to 500 ms. The rate of the MIP population was set to 20 Hz to illustrate that even at low firing rates, spike correlations [$c = 0.5$, e.g. (see Fig. 3*d*)] could induce strong transients.

Data analysis

Event characterization

To characterize the spiking activity of each group and, thus, the α and σ of the pulse packet sent to the next group, we measured "events," i.e., distinct, short bursts of spikes in the peristimulus time histogram (PSTH) of all excitatory neurons in a group. The spike count of all excitatory neurons in an event described the response strength α. Likewise, σ described the temporal spread of the spike times, that is, the time difference between the start and end of the "event." Thus, the propagation of activity in the signal path was determined by the response of the three groups of the signal path, quantified by α and σ.

Signal propagation

To determine whether a signal, composed of multiple pulse packets, could successfully propagate from the sender to the receiver, we measured the similarity between the spiking activities in both groups. The similarity was calculated by cross-correlating the PSTHs of the excitatory neurons in both groups. In case of successful propagation, the signal in the receiver group should have a similar temporal sequence of peaks in the PSTH. However, due to the synaptic delay in the signal path, this sequence would be shifted in time. Therefore, when calculating the cross-correlation between sender and receiver, we accounted for this time shift.

Simulation tools

All network simulations were written in Python (http://www.python.org) using PyNN (Davison et al., 2008) (http://neuralensemble.org/trac/PyNN) as interface to the simulation environment NEST (Morrison et al., 2005; Gewaltig and Diesmann, 2007; Eppler et al., 2008) (http://www.nest-initiative.org). The dynamical equations were integrated at a fixed temporal resolution of 0.1 ms. Simulation management was performed using the Python package NeuroTools (http://neuralensemble.org/trac/NeuroTools). Simulation data were analyzed in Python using the scientific libraries SciPy (http://www.scipy.org) and NumPy (http://numpy.scipy.org) and visualized using the plotting library Matplotlib (http://matplotlib.sourceforge.net).

Results

To systematically study the concept of temporal gating, we embedded a signal path in a large-scale recurrent cortical network model (Fig. 1*a*,*b*). The signal path was a feedforward network consisting of three successive groups of neurons. The first group, composed of excitatory neurons activated by the stimulus, is referred to as the sender. The second and third groups contained both excitatory and inhibitory neurons, with the inhibitory neurons providing disynaptic inhibition to the excitatory neurons

E15 局部连接的空间核
作者报告｜连接概率由高斯核给出，兴奋与抑制的空间常数不同，为空间分组的门控提供结构基础。

E22 脉冲包刺激的规模
作者报告｜脉冲包刺激与通路内连接使用同一规模，使 σ 的取值有明确的物理含义。

E16 外部输入强度与工作点
作者报告｜外部输入被调到低发放、异步不规则态，是本文所依赖的生理工作点设定。

E23 相关 rate 刺激的参数
作者报告｜20 Hz 加 0.5 复制概率的设计意在说明低发放率下相关性也能造成强瞬变，这是对「平均发放率不足以描述信号」的直接回应。

E17 抑制神经元在通路中的作用
作者报告｜作者把抑制神经元的双重作用写在方法里：既提供平衡抑制，又决定 τ_eff——后者正是本文门控的施力点。

E24 传播判据：PSTH 互相关
作者报告｜信号是否成功传播用 sender 与 receiver 的 PSTH 互相关判定，并按通路延迟做了时间平移，因此指标衡量的是时间结构是否重现。

E18 通路内的汇聚输入
作者报告｜组间汇聚输入的规模决定了单次刺激能驱动多少下游神经元，是 σ 的物理来源。

E19 通路内兴奋延迟的自述任意性
作者报告｜作者主动交代参数的任意性，是方法透明度上的加分项，也提示读者该延迟不是机制本身。

Ｅ２０　　作者对该延迟无关性的论证
作者报告｜作者论证兴奋延迟可取任意值而不影响门控，但这是作者声明，本文未做扫描验证。

Ｅ２１　　默认抑制延迟
作者报告｜２　ｍｓ　的默认抑制延迟就是本文全部扫描的基线，门控效果都相对它定义。

Ｅ１０　　研究对象的整体架构
作者报告｜说明信号通路是嵌入到既有随机网络中的实验对象，背景活动因此不是被简化掉的理想输入。

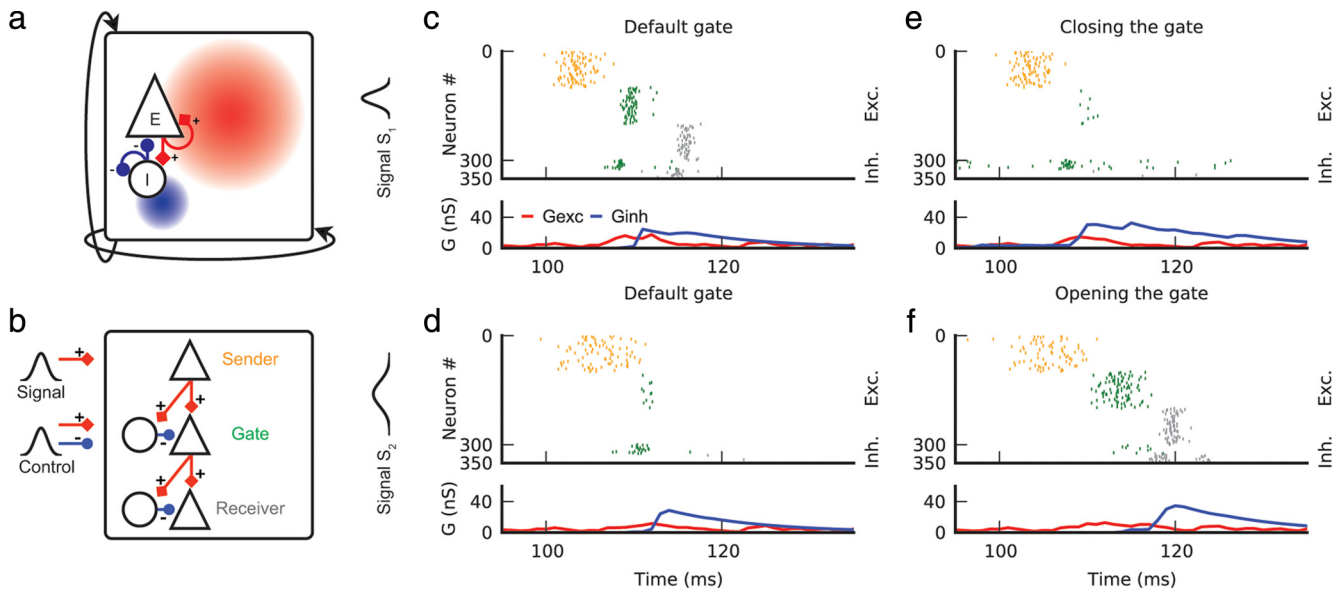


Figure 1. Temporal gating of spiking activity by delayed correlation between excitation and inhibition. ***a***, Scheme of cortical network model. Twenty-eight thousand, one hundred twenty-five neurons [80% excitatory (△) and 20% inhibitory (○)] were arranged as a two-dimensional sheet. Space constants of excitatory and inhibitor neurons are shown as red and blue shaded regions, respectively. ***b***, A signal path (sender, gate, receiver) was embedded in the network. The sender neuron group transmitted an input signal via the gate group toward the receiver group. A depolarizing or hyperpolarizing input to the gate was used as control signal. ***c***, ***d***, Top, Spiking activity of the sender, gate, and receiver groups. Bottom, Excitatory and inhibitory synaptic conductances (Gexc, Ginh) for a representative excitatory neuron in the gate group. ***c***, A strong ($\alpha = 60$), synchronous ($\sigma = 3.5$ ms) signal S_1 propagated through the gating group and successfully activated the receiver group. ***d***, By contrast, a less synchronous ($\sigma = 7$ ms) signal S_2 failed to propagate. ***e***, Propagation of S_1 could be blocked by advancing inhibitory spiking in the gate group, thereby reducing Δt of the excitatory neurons. ***f***, Propagation of S_2 could be rescued by delaying (or abolishing) inhibitory spiking in the gate group, thereby increasing Δt of the excitatory neurons.

within the same group. The third group is referred to as receiver, whereas the second group, which is in a strategic position to influence propagation from sender to receiver, is referred to as the gate (Fig. 1*b*). However, the gating mechanism is flexible and in a feedforward network with more than three groups, any group can be considered as a gate group between its preceding and succeeding partner groups. The embedding network was tuned into an asynchronous-irregular state to provide realistic background activity (Kumar et al., 2008a) (cf. Materials and Methods).

Temporal gating of signal propagation

To illustrate the mechanism of temporal gating, we stimulated the sender group with a volley of spikes (pulse packet; cf. Materials and Methods), with α spike times drawn from a Gaussian distribution of width σ.

We first considered the propagation of a synchronous stimulus S_1 (small σ). In the default state of the signal path (cf. Materials and Methods), S_1 generated a transient response in the sender, which successfully propagated through the gate to the receiver (Fig. 1*c*). This propagation can be described in terms of the synfire attractor, a defining feature of feedforward network dynamics (Diesmann et al., 1999; Kumar et al., 2008a, 2010; Kremkow et al., 2010), with the fate of signal propagation being determined by the properties of the pulse packet stimulus. The (α, σ) state space is subdivided by a separatrix (schematically shown in Fig. 2*a*, blue line). If the pulse packet starts above the separatrix (class *P* stimulus), the signal converges into a fixed point (FP) and propagates in a stable manner (Figs. 1*c*, 2*a*). If, by contrast, the pulse packet starts below the separatrix (class *F* stimulus), the signal progressively dies out, indicating failure to propagate (Figs. 1*d*, 2*a*). Thus, the location of the separatrix is the critical parameter that controls the propagation of a stimulus (Diesmann et al., 1999; Kumar et al., 2008a, 2010; Kremkow et al., 2010). Consequently, altering the location of the separatrix could change a class *P* stimulus into a class *F* stimulus (or vice versa), thereby providing a simple mechanism for gating the propagation of more or less transient spiking activity (Fig. 2*a*). Various properties of the feedforward network, such as group size, connection probability, and weights determine the location of the separatrix in the (α, σ) space, which, however, can also be modulated by the dynamical properties of the feedforward network, such as noise level of the embedding background activity (Diesmann et al., 1999; Gewaltig et al., 2001). Furthermore, we recently showed that the delay between excitation and inhibition within one group changes the slope of the separatrix by effectively altering the integration time (Δt) of the neurons (Kremkow et al., 2010). Interestingly, recent data from *in vivo* measurements of the delay between excitation and inhibition (Okun and Lampl, 2008) are in the right range to alter the slope of the separatrix in an appreciable way. Several biologically plausible mechanisms could lead to dynamic variations in the excitation/inhibition delay and, thereby, to associated modulations of Δt. For instance, neuromodulators could change Δt by differentially altering neuronal excitability or synaptic strengths (Kruglikov and Rudy, 2008; Antal et al., 2010). Similarly, stimulus properties could introduce variations in Δt (Zhang et al., 2003; Wilent and Contreras, 2005).

We exploited the control over the delay between excitation/inhibition to close the gate, i.e., to block the propagation of the S_1-type stimulus as shown in Figure 1*c*. Thus, we reduced the temporal delay between excitation and inhibition in the gating group by injecting a small depolarizing control pulse into the inhibitory neurons. As a result, the gate closed: the response in the gate group fell below the separatrix, causing activity in the receiver group to fade (Figs. 1*e*, 2*a*). To further explore the temporal gating mechanism, we systematically varied Δt by altering the inhibitory synaptic delay, while keeping the excitatory delay fixed. For different values of Δt, Figure 2*b* shows α and σ for

E25 关门操作
作者报告｜关门被实现为对抑制神经元注入去极化脉冲以提前其发放，说明本文的门控旋钮是延迟而非权重。

E26 关门的结果
作者报告｜关门与分界线下降直接对应，说明作用点确实是同步活动状态空间的分界线而非单纯的活动抑制。

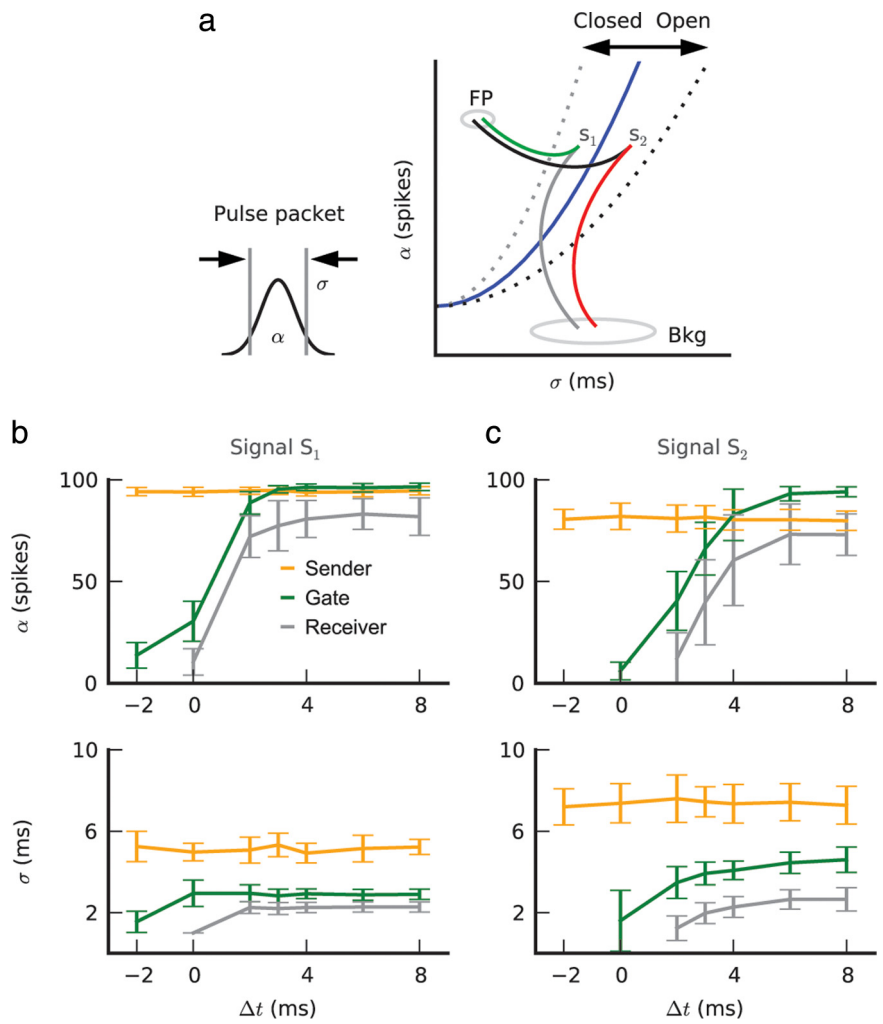


Figure 2. Temporal delay between excitation and inhibition controls signal gating. ***a***, Scheme of the pulse packet propagation state space, which is subdivided by a separatrix (Diesmann et al., 1999) (blue line). If the pulse packet starts above the separatrix (S_1), the signal converges into a fixed point (FP) and propagates in a stable manner (green trajectory). If, by contrast, the pulse packet starts below the separatrix (S_2), the signal fails to propagate and fades into the background (Bkg.) (red trajectory). When the effective integration time Δt is reduced, the separatrix moves up (gray dotted line), thus blocking the propagation of S_1 (gray trajectory). By contrast, when the integration time is increased, the separatrix moves down (black dotted line), thus allowing the propagation of S_2 (black trajectory). ***b***, α and σ in sender, gate, and receiver groups for the S_1 stimulus (class *P*) as a function of effective integration time (Δt). Error bars represent SD ($n = 20$). In this example, S_1 successfully propagates in the default state ($\Delta t = 2$ ms). Further increase in Δt does not influence signal propagation. However, a small decrease in Δt (~1 ms) suffices to completely block signal propagation. ***c***, Same as ***b*** for stimulus S_2 (class *F*). In this example, S_2 does not propagate in the default state ($\Delta t = 2$ ms). Propagation can be rescued by increasing Δt. Note that, in contrast to ***b***, the prolonged opening of the gate causes activity in the gate to be higher than in the receiver. Also, for increasing integration time (larger Δt), signal propagation becomes more reliable, which is expressed as a reduction of trial-by-trial variability in the receiver group.

the three excitatory neuron groups in the signal path. For small integration time ($\Delta t \leq 1$ ms in this example), the gate response was weak and fell below the separatrix. Consequently, the response in the receiver group was indistinguishable from the background network activity. However, increasing the integration time ($\Delta t \approx 2$ ms in this example) shifted the gate activity above the separatrix, inducing a strong response (high α, small σ) in the receiver group. The same mechanism can be used to open the gate, i.e., to facilitate propagation of class *F* stimuli, which would otherwise fail to propagate (Fig. 1*f*). To test this, we applied a hyperpolarizing (rather than depolarizing) control pulse to the inhibitory neurons, thereby delaying or even suppressing their spiking activity. As a result, the integration time was increased, such that more excitatory neurons in the gate group responded. Its activity thereby rose above the separatrix and, hence, ensured a strong response (high α, small σ) in the receiver group (Fig. 1*f*). A systematic increase in the integration time Δt of the gate neurons resulted in a monotonic increase in both α and σ of the gate activity (Fig. 2*c*). Due to the synfire attractor, the activity of the receiver group was further synchronized. Note that, in contrast to Figure 2*b*, the prolonged opening of the gate caused activity in the gate to be higher than in the receiver. Also, it is noteworthy that for increasing integration time (higher Δt), signal propagation became more reliable, as expressed in a reduction in trial-by-trial variability in the receiver group (Fig. 2*c*, top panel).

In summary, the sigmoidal relationship between Δt and the activity in the gate and receiver groups demonstrates the effectiveness of the temporal gating (Fig. 2*b*,*c*).

As there is an ongoing debate (Kuhn et al., 2004; Vogels and Abbott, 2005; Kumar et al., 2008a,b) on whether synapses in a point neuron model are best described by conductance transients (as done here) or current transients [the more common choice, e.g., Brunel (2000)], we resimulated the network model with current-based synapses. The sigmoidal relationship between Δt and the activity in the gate and receiver groups remained unchanged when synapses were modeled as current transients (compare supplemental Fig. S1, available at www.jneurosci.org as supplemental material, and Fig. 2*b*,*c*), demonstrating that the "temporal gating" does not depend on the choice of synapse or input activity model.

Gating of signals with strong transients

Next we tested whether temporal gating is also effective in controlling the propagation of signal onset transients. Such signal onset transients arise when the input signal is abruptly changed (Deger et al., 2009). To mimic sharp changes in the input, the sender group, we stimulated the sender group with Poisson-type spiking activity, starting at time $t = 500$ ms (Fig. 3*a*). In the default state of the gate group, with balanced excitation and inhibition, the onset transient propagated to the receiver group, whereas the tonic part of the input was blocked (Fig. 3*b*). The control of such transients, however, is critical because in a dynamic natural environment, such transients are likely to occur frequently (Buracas et al., 1998; Haider et al., 2010). In the default state of the gate, the effective integration time is large enough for the transients to elicit a response in the downstream neurons (gate group). Thus, an effective way to control the flow of transients is to reduce the integration time of the downstream neurons (e.g., by temporal gating) (see Figs. 1, 2). Indeed, when the delay between excitation and inhibition in the gate group was reduced, while still keeping the overall excitation and inhibition in balance, it was possible to control the propagation of both the

E30 门控关系的 S 形性质
作者报告｜积分时间增加时 gate 组活动的强度与时间宽度都单调增加，说明 τ_eff 不是二值开关而是连续旋钮。

E31 可靠性随门开程度提高
作者报告｜τ_eff 增大不仅恢复幅度还降低试次间变异，是超出「能通／不能通」的额外证据，也是 Fig. 2 有 n = 20 误差条的部分。

E32 电流基突触对照
作者报告｜换成电流基突触后 S 形关系不变，据此声称机制不依赖突触模型选择；该结论依赖未随本文提供的补充材料 Fig. S1。

E33 对模型选择的无关性
作者报告｜作者把这条提升为一般性结论。用在模型中时合理，但引用时应注明其证据来自未核对的补充材料。

E34 平衡状态下瞬变漏过
作者报告｜本文批评 P2 的核心观察：同一条件下持续成分被挡住而 onset 瞬变通过，直接说明只测时间平均放电会漏判。

E28 阻断的定量阈值
作者报告｜积分时间降到 1 ms 以下时 gate 响应跌破分界线，receiver 组响应与背景活动不可区分，给出了关门的定量阈值。

E29 开门后的响应
作者报告｜关闭时 receiver 输出退回背景，说明门控是全或无式的传播失败，而不是把信号按比例削弱。

E27 开门操作
作者报告｜开门是关门的反向操作，两个方向都有对应实验，使 τ_eff 的因果作用比单方向扫描更难被替代解释推翻。

stimulus transient and its tonic component (Fig. 3*c*). Note that in this particular example, when the gate is closed (Fig. 3*c*), only the inhibitory population in the gate group is active, while the activity of the excitatory neurons is completely suppressed by the balancing inhibitory input. Reducing the gain of the inhibitory neurons would result in a biologically more realistic residual spiking activity in the excitatory neurons in the gate group. Transients in the spiking activity in neural systems may also arise due to spike time correlations among neurons (Alonso et al., 1996; Dan et al., 1998; Chen et al., 2006). The emergence of such correlations could be due to large-scale fluctuations in ongoing network activity (Arieli et al., 1996; Kenet et al., 2003) or stimulus-induced network activity (Bar-Yosef and Nelken, 2007; Alonso et al., 2008; Hromádka et al., 2008). To generate a signal composed of tonic and transient components, we used the MIP model (Kuhn et al., 2003) (Fig. 3*d*; cf. Materials and Methods). Once again, when the sender group was stimulated by a MIP-type input signal, the control of the activity was imprecise and signal transients reflecting the spike correlations propagated uninhibited (Fig. 3*e*). However, reducing the delay between excitation and inhibition at the gate group provided an effective control over the propagation of signal transients (Fig. 3*f*). In summary, Figure 3, *b* and *e*, illustrates the limitations of the Vogels and Abbott mechanism in gating of both onset transients (Fig. 3*b*) and transients induced by correlations (Fig. 3*e*). Both types of transients were fast enough to pass before the delayed feedforward inhibition could quench them. By contrast, Figure 3, *c* and *f*, shows the improved effectiveness of the temporal gating mechanism in controlling the propagation of these activity transients.

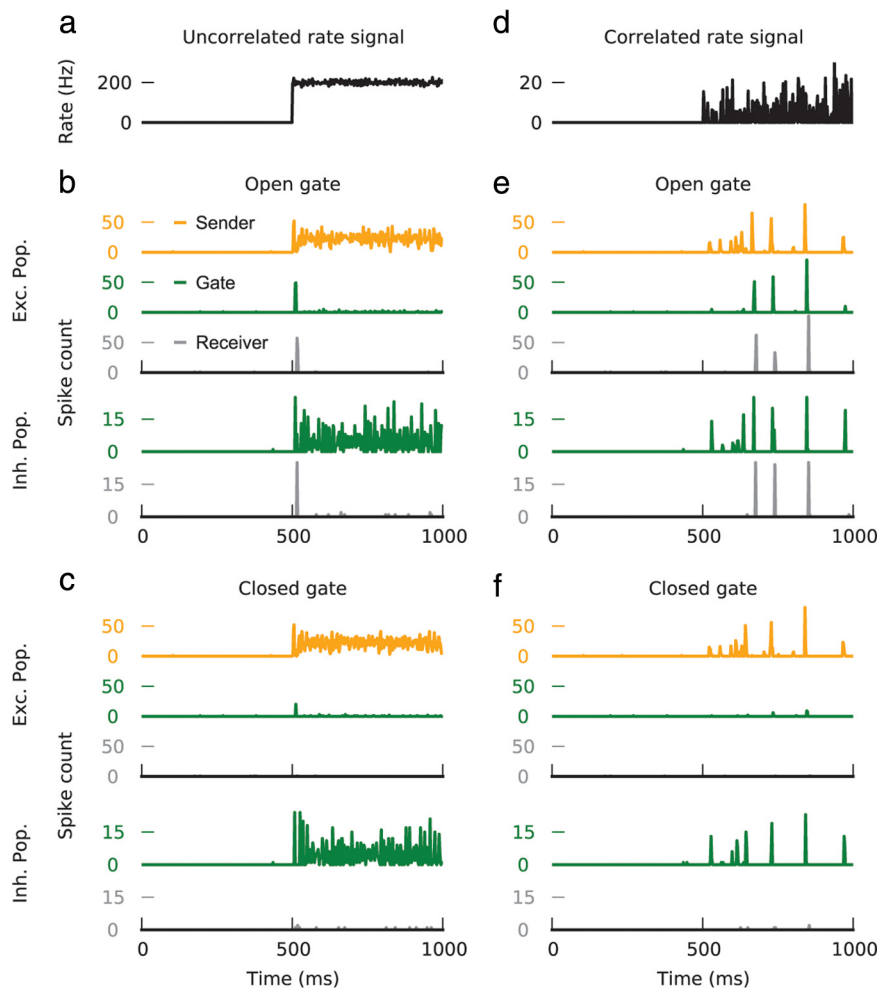


Figure 3. Effectiveness of gating tonic activity and activity transients. ***a***, Uncorrelated rate stimulus as input to the signal path. ***b***, Population activity in the excitatory (top) and inhibitory (bottom) neurons in the signal path. In the default state ($\Delta t = 2$ ms), gating by balancing excitation and inhibition successfully blocks propagation of the tonic component of the stimulus, but fails to block the stimulus onset transient. ***c***, By reducing the effective integration time of the excitatory neurons ($\Delta t \leq 1$ ms), temporal gating effectively blocks both the onset transient and the tonic stimulus component. ***d***, Firing rate stimulus with transient input correlations. ***e***, As in ***b***, gating by balancing excitation and inhibition fails to block activity transients, here induced by transient input correlations. ***f***, As in ***c***, temporal gating effectively blocks activity transients.

Combining temporal and amplitude gating

The effectiveness of temporal gating in controlling the propagation of activity transients has its price: it precludes the propagation of sustained (tonic) activity devoid of transients (Fig. 3*c*,*f*). While the importance of such sustained average neural activity can be argued, in view of the ubiquitous fluctuations in the ongoing activity and dynamic natural environment, it is of general interest to know whether it is possible to do both, to control the flow of activity transients, while allowing tonic activity to propagate. Because temporal gating is most effective in controlling the propagation of activity transients, whereas detailed balance (or amplitude gating) can be used to control the propagation of tonic activity, a combination of the two might provide a general mechanism to control the propagation of arbitrary neural activity patterns. Therefore, we examined whether a combination of temporal gating and amplitude gating would be possible within the same network.

For this we stimulated the signal path with uncorrelated Poisson activity, starting at time $t = 500$ ms. This type of stimulus contains both a transient onset and a tonic phase, which makes it optimal for probing the effectiveness of both gating mechanisms. The amplitude gating mechanism was implemented by varying the strength of the inhibition in the gate group, thereby controlling the balance of excitation and inhibition in the excitatory neurons of the gate group (Vogels and Abbott, 2009). To this end, we varied the synaptic strength from the excitatory neurons in the sender group onto the inhibitory population (J_{EI}), while keeping the synaptic strength from the excitatory neurons onto the excitatory neurons (J_{EE}) fixed. Thus, we define the inhibitory gain factor as J_{EI}/J_{EE}. In all examples shown so far (i.e., Figs. 1–3), the inhibitory gain factor was 2. This gain factor ensured an effective feedforward inhibition in the neurons of the gate group (cf. Materials and Methods) (Cruikshank et al., 2007; Kremkow et al., 2010). Figure 4 shows the results of systematically varying both the degree of temporal gating and amplitude gating. Simultaneous closing of both the temporal ($\Delta t = -2$ ms) and amplitude (inhibitory gain = 2) gates blocked both the transient and the tonic component in the gate group (Fig. 4*a*–*c*). Opening the tem-

E39 关门时的实现细节
作者报告｜门关闭时 gate 组只有抑制群体在活动、兴奋群体被完全压制，作者自己认为这在生物学上不够真实。

E35 相关输入的瞬变同样漏过
作者报告｜低发放率下由相关性产生的瞬变也未被控制，说明缺陷不限于高幅值刺激。

E36 时序门控对两类瞬变都有效
作者报告｜缩短 E/I 延迟同时挡住了 onset 与相关两类瞬变，是时序门控相对幅度门控的直接优势证据。

E37 幅度门控失效的机理解释
作者报告｜给出了漏过的机制解释——瞬变快于抑制窗口的形成时间，因此引用本文批评 P2 时应连同这个限定一起引用。

E40 幅度门控的实现方式
作者报告｜幅度门控通过改 J_EI 实现，与 P1 中「改增益等于同等放大全部突触」的操作并非同一件事。

E38 关门的代价
作者报告｜作者自陈时间门控不能传递无瞬变的持续活动，这是本文最重要的适用边界，也是它必须与幅度门控合用的原因。

E41 抑制增益因子的定义
作者报告｜把 J_EI/J_EE 定义为单一的抑制增益因子，使两个门控变量可以在同一平面上正交扫描。

E42 默认增益因子为 2
作者报告｜Fig. 1 至 Fig. 3 全部使用同一增益因子，说明早期结果是在单点配置下取得的，范围性有限。

E43 两门同时关闭
作者报告｜时间门与幅度门同时关闭时两个成分都被挡住，说明两种门控在同一条件下可以叠加生效。

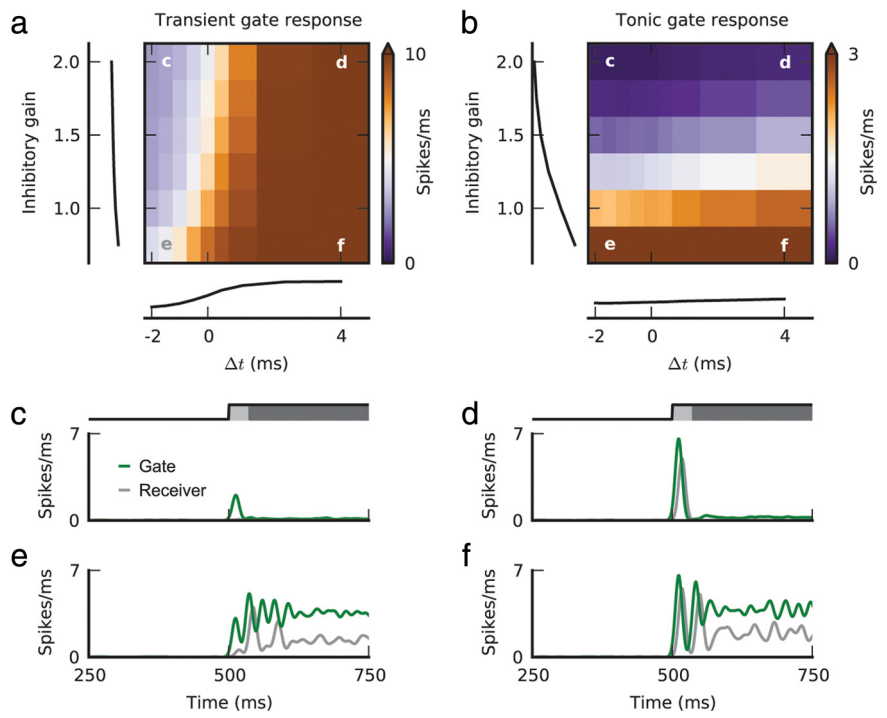


Figure 4. Combining temporal and amplitude gating. ***a***, Transient response (first 10 ms, marked light gray in ***c*** and ***d***) in the gate as a function of effective integration time Δt (temporal gating) and inhibitory gain (amplitude gating) with uncorrelated rate stimulus as input to the signal path. Mean transient response in the gate does not change as a function of inhibitory gain for all values of Δt, indicating that inhibitory gain cannot gate the propagation of transient component of the activity (left). By contrast, mean transient response in the gate as a function of Δt for all values of inhibitory gain shows a sigmoidal shape, indicating that Δt can control the propagation of transient component of the activity (bottom). ***b***, Tonic response (excluding first 10 ms, marked dark gray in ***c*** and ***d***) in the gate as a function of effective integration time Δt and inhibitory gain. Mean tonic response in the gate as a function of inhibitory gain for all values of Δt, varies in a threshold linear manner, indicating that inhibitory gain can control the propagation of tonic component of the activity (left). By contrast, mean tonic response in the gate does not change as a function of Δt for all values of inhibitory gain, indicating that Δt cannot gate the tonic component of the activity (bottom). ***c–f***, Population activity of the excitatory neurons in the gate and receiver group during four distinct gating modes, marked with corresponding letters in ***a*** and ***b***. ***c***, Simultaneously closing of the temporal and the amplitude gates blocks both the onset transient and the tonic component, resulting in a very small activity in the receiver group. ***d***, Opening the temporal gate, while keeping the amplitude gate closed, allows the onset transient to propagate. ***e***, Opening the amplitude gate, while keeping the temporal gate closed, allows the tonic component to propagate. ***f***, Both the transient and the tonic component are propagated when both gates are open.

poral gate ($\Delta t = 4$ ms), while keeping the amplitude gate closed (inhibitory gain = 2), allowed only the onset transient to induce activity in the gate group (Fig. 4*a*,*b*,*d*). These two examples, thus, show the effectivity of temporal gating at high inhibitory gain (compare Figs. 3 and 4*c*,*d*). By contrast, opening the amplitude gate (e.g., inhibitory gain = 0.75) resulted in tonic activity in the gate group (Fig. 4*a*,*b*,*e*,*f*). Here, the weaker feedforward inhibition could not balance the direct excitation during the tonic phase (Kremkow et al., 2010). However, during the transient phase the inputs from the sender group arrived synchronously, activating the inhibitory neurons in the gate group simultaneously, thereby providing sufficient balancing inhibition for the temporal gating to function and block the transient [compare closed temporal gate (Fig. 4*e*) and open temporal gate (Fig. 4*f*)]. Note that using inhibitory gain as a gating variable may result in an attenuated response in the receiver group. However, because a high signal-to-noise ratio is maintained, this reduced activity could easily be amplified in a subsequent stage.

In summary, Figure 4 shows that amplitude gating is effective in gating tonic component of the activity (Fig. 4*b*), while temporal gating is effective in gating the activity transients (Fig. 4*a*). When the two gating mechanisms are implemented within the same network, together they provide a powerful mechanism to control the propagation of a wide variety of temporal activity profiles.

Gating of multiple signals

So far we considered how isolated transients or those that form part of stimulus-evoked or ongoing activity modulations propagate along a single signal path. However, in a modular system like the brain, action selection can be thought of in terms of directing one of multiple signals to a next processing stage or, alternatively, as directing a particular signal to one of multiple subsequent processing stages (Bienenstock, 1995). In both these scenarios, the brain needs a mechanism to control the propagation of activity over multiple (and interacting) signal pathways. Here, we test the effectiveness of temporal gating in selecting between two neuronal activity signals that are statistically identical but uncorrelated, and dominated by transients.

Figure 5*a* shows the scheme of two signal paths, embedded in a recurrent random network (cf. Materials and Methods). This arrangement of sender, gate, and receiver groups is similar to that used in Figures 1–4, except that here two spatially separated sender groups (Sender_X and Sender_Y) can send signals (S_X, S_Y) to the gate. To control the propagation of either signal S_X or S_Y, we systematically varied the delay between excitation and inhibition from the respective sender to the gate (Δt_X and Δt_Y, respectively). To quantify the propagation of the two signals, we measured the cross-correlations (ρ_X and ρ_Y) between the receiver group output and each of the two input signals (cf. Materials and Methods). Figure 5*b* shows the difference between the two: $\rho = \rho_X - \rho_Y$ as a function of both Δt_X and Δt_Y. Positive values of ρ indicate that the receiver group output is more similar to the input signal S_X, and negative values reflect a higher similarity with input signal S_Y. When ρ is close to zero, the receiver output is equally correlated (including uncorrelated) with both input signals; in this case, the scenario is evidently not suitable for action selection.

When the two signals were presented simultaneously, selective transmission of signal S_X (Fig. 5*b*,*c*) could be obtained by increasing Δt_X and decreasing Δt_Y (effectively closing signal path S_Y) (compare Fig. 2). Similarly, selective transmission of signal S_Y (Fig. 5*b*,*f*) could be obtained by increasing Δt_Y and decreasing Δt_X (closing signal path S_X). The exact value of Δt_X to gate signal S_X depended on the value of Δt_Y (Fig. 5*b*) and vice versa. When the gate was closed for both signal paths (Fig. 4*e*), neither of the two input signals propagated to the receiver and, hence, ρ was close to zero. On the other hand, when the gate was opened for both signal paths (Fig. 5*d*), both signals propagated and the response of the receiver group was equally correlated to both input signals, again resulting in ρ being close to zero.

Thus, the temporal gating mechanism appears well suited for selective gating of multiple signals to a next processing stage.

E48 两条通路的布局
作者报告｜选择性实验建立在两条并行信号通路上，用与 Fig. 1 至 Fig. 4 相同的 sender/gate/receiver 结构。

E49 按通路分别调节 τ_eff
作者报告｜两条通路的 E/I 延迟被独立调节，是选择性能够出现的直接原因。

E50 选择性指标的构造
作者报告｜选择性由 receiver 输出与两输入信号的互相关之差刻画，是一个差值型指标。

E44 只开时间门
作者报告｜只开时间门时只有 onset 瞬变通过、持续成分被挡住，直接对应两个成分的分工。

E51 选择性方向的可解释性
作者报告｜指标符号可解读为「更像哪个输入」，因此正负号可以当作选择方向的判据。

E53 作者对 Δσ 失效区的说明
作者报告｜作者明确承认 Δσ 接近零意味着场景不适于动作选择，引用该指标时必须带上这一自我限定。

Ｅ４５　只开幅度门时瞬变仍被挡住
作者报告｜说明幅度门打开后持续成分通过，而瞬变期因输入同步激活抑制群体仍被时间门控挡住，两个成分确实由不同变量控制。

Ｅ５４　选择性传播的实现方式
作者报告｜增大一条通路的 τ_ｅｆｆ、同时减小另一条即可实现选择性传播，说明控制量可以按通路分别设定。

Ｅ４６　瞬变期抑制被同步激活
作者报告｜给出两门合用时瞬变仍被挡住的机制解释：瞬变的同步性本身召回了抑制，使时间门控得以生效。

Ｅ５５　　两条门控状态的相互依赖
作者报告｜反向配置实现对　Ｓ＿Ｙ　的选择性；结合作者关于阈值相互依赖的说明，两个门控状态并非独立叠加，这一依赖机制未被展开。

Ｅ４７　　幅度门控的副作用与作者的处理
作者报告｜作者承认用抑制增益作门控变量会让　ｒｅｃｅｉｖｅｒ　响应衰减，并认为在高信噪比下可在后续级放大，是一条需要检验的乐观判断。

Ｅ５２　　两门都开时的度量塌缩
作者报告｜两门同开时输出对两个输入同等相关，Δσ　因此趋零，说明该指标无法区分全通与全断，是使用时最容易误读的一处。

Similarly, temporal gating can also be used to selectively direct a particular signal to one of several next processing stages (not shown here).

Discussion

In a modular system like the brain, appropriate gating or control over activity propagation is critical for information processing and action selection (Bienenstock, 1995; Kumar et al., 2010). In fact, for various biological reasons, gating of neural activity emerges as a superior strategy over winner-take-all dynamics for action selection (Ermentrout, 1992; Tepper et al., 2004). In any case, a key requirement for an appropriate gating mechanism is that it should be able to control neural activity with arbitrary statistical properties.

In particular, it is important to be able to gate neuronal activity that is dominated by strong transients, which are ubiquitous in a dynamic sensory environment, as has been reported under various conditions *in vivo* in all sensory pathways. For example, it has been demonstrated in the rodent whisker-barrel system that cortex is primarily driven by transient synchronous activity of thalamic neurons (Bruno and Sakmann, 2006). Similarly, neurons with overlapping receptive fields of the same polarity (on- or off-center) in the cat visual thalamus show spike correlation in the millisecond range and, consequently, drive postsynaptic neurons in the visual cortex much more effectively (Alonso et al., 1996). Furthermore, transient subthreshold dynamics have been measured intracellularly in the cat visual cortex *in vivo* during presentation of natural stimuli (Haider et al., 2010; Y. Fregnac, personal communication). Likewise, subthreshold membrane potential *in vivo* in the rat auditory cortex showed large transients, induced by synchronous synaptic input, occurring irregularly in an otherwise quiet background (DeWeese and Zador, 2006). Together, these observations in different sensory pathways in several species corroborate the importance of activity transients. Consequently, a proper gating mechanism should be able to cope with these transients.

Recently, Vogels and Abbott (2009) proposed a gating mechanism based on the detailed balance of excitation and inhibition to control the propagation of neuronal activity in feedforward networks. This gating strategy, however, is only effective in controlling the propagation of tonic components of neuronal activity modulation—transients "escape" from such gating and propagate to the next stage in any case. This is because the mechanism fails in gating transients that are fast enough to pass the small time window, induced by the lagging inhibition, i.e., in the range of some tens of milliseconds [Vogels and Abbott (2009), their Fig. 3*f*]. As a consequence, gating based only on detailed balance of excitation and inhibition is truly limited in its applicability.

The temporal gating mechanism proposed here extends this idea by changing the delay between excitation and inhibition, and thereby obtaining the means to control the temporal scale of transients that can pass or, alternatively, are blocked. The delay between excitation and inhibition affects the neurons integration time (Pouille and Scanziani, 2001; Cruikshank et al., 2007). Thereby, it can be used to block or facilitate activity propagation. Thus, this gating mechanism presents an example of exploiting a single neuron-level property to modulate network behavior—in this case, the location of the state space separatrix—such as to control the propagation of spiking activity within the network. As our study showed, the temporal gating mechanism is a simple yet powerful mechanism to control the propagation of activity transients. Thus it is complementary to the amplitude gating mechanism proposed by Vogels and Abbott (2009). When the two mechanisms are combined, they can control neural activity with different temporal profiles.

The key parameter of temporal gating is the delay between excitation and inhibition. The delayed correlated inhibition in our model was induced by disynaptic inhibition, a connectivity pattern observed ubiquitously in the CNS [hippocampus (Buzsáki, 1984), thalamocortical connection (Swadlow, 2003; Cruikshank et al., 2007), local cortical network (Kapfer et al., 2007; Silberberg and Markram, 2007), and long-range horizontal cortical network (Hirsch and Gilbert, 1991; Tucker and Katz, 2003a,b)]. In addition, neuromodulators (Kruglikov and Rudy,

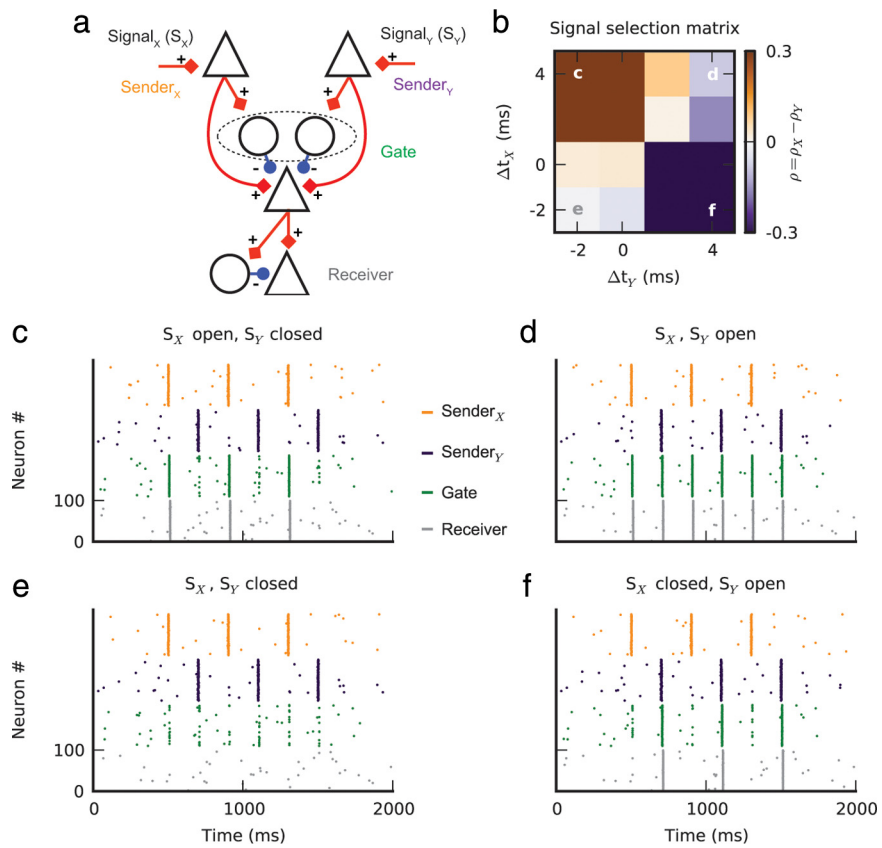


Figure 5. Selective gating of multiple signals. ***a***, Scheme of two interacting signal paths. ***b***, Selectivity of signal propagation in two interacting signal paths. The color map shows the selectivity measure $\rho = \rho_X - \rho_Y$ as a function of effective integration time Δt_X and Δt_Y. Positive values (brown color) of ρ indicate that the receiver group output is more similar to signal S_X, and negative values (blue color) reflect a high similarity with signal S_Y (i.e., selective gating of either S_X or S_Y). At $\rho \approx 0$, the receiver output is equally correlated with both signals (i.e., no selective gating). ***c–f***, Spiking activity of the sender, gate, and receiver groups during four distinct gating modes, marked with corresponding letters in ***b***. ***c***, With the gate open for S_X and closed for S_Y, the activity in the receiver group resembles the signal S_X. ***d***, With both gates open, the activity in the receiver group is a superposition of S_X and S_Y, thereby reducing ρ to zero. ***e***, Closing both gates blocks the propagation of both signals. ***f***, With the gate open for S_Y and closed for S_X, the activity in the receiver group resembles the signal S_Y.

E56 对前作适用性的总体判断
作者报告｜作者把批评落在成分层面而非否定机制整体，引用时应与「幅度门控完全无效」的误读区分开。

E59 硬件的通用性
作者报告｜延迟抑制由双突触连接实现，作者称该连接模式在 CNS 普遍存在，是机制生物学合理性的主要论据。

E57 漏过的机制层级解释
作者报告｜把失效归因于抑制滞后造成的时间窗，并给出几十毫秒的量级，是讨论部分对结果的最强机制表述。

E58 机制定位：单神经元属性调制网络行为
作者报告｜作者把机制定位为用单神经元的积分时间属性去移动网络级分界线，这正是把它作为可实现模型元件的理由。

2008; Antal et al., 2010) and stimulus properties (Zhang et al., 2003; Wilent and Contreras, 2005) could change the latency between excitation and inhibition in a neuron specific manner, by differentially altering neuronal excitability or synaptic strengths. Thus, the required neuronal hardware for the proposed temporal gating mechanism is quite general and not restricted to any specific brain area. Furthermore, the gating mechanism does not depend on the choice of the synapse model (current based vs conductance based) (Fig. 2 and supplemental Fig. S1, available at www.jneurosci.org as supplemental material). Moreover, the delay values used here were well within the range measured experimentally *in vivo* (Okun and Lampl, 2008). The effectivity of temporal gating in controlling the propagation of activity transients, however, precludes propagation of sustained (tonic) neural activity devoid of transients (Fig. 3*c*,*f*). Because detailed balance (or amplitude gating) is suitable in controlling the propagation of tonic activity, a combination of the two mechanisms provides a general solution to control the propagation of arbitrary neural activity patterns.

Here we presented a possibility of combining the two mechanisms and showed that, indeed, by comodulating excitation–inhibition balance and their latencies, it is possible to control the propagation of neural activity for combinations of transient and tonic components (Fig. 4). In addition, more sophisticated mechanisms might be invoked. For instance, dynamic modulation of excitation–inhibition balance can be achieved with depressing or facilitating synapses. Thus, introducing synapses with short-term plasticity (Klyachko and Stevens, 2006) into our temporal gating mechanism might be an alternative way of combining the strengths of the two gating mechanisms.

Gating of transients is most effective when the activity transients are close to the separatrix (Fig. 2*a*). If, by contrast, the activity transients are close to the stable fixed point (FP) (Fig. 2*a*) or too similar to the background fluctuations (Bkg) (Fig. 2*a*), temporal gating may not be as effective. Thus, if temporal coding is indeed used in neuronal systems, it should be possible to predict the spectrum of neural activity transients, given the properties of synapses and the connection probabilities between processing stages. In a similar vein, knowledge of the neural activity transients may provide important hints concerning the connectivity between subsequent processing stages.

Preliminary experimental evidence already points toward the involvement of a temporal gating mechanism in decision making and action selection. For instance, animals can make decisions on a millisecond timescale (Yang et al., 2008). Mechanisms based on winner-take-all dynamics would not support such rapid decision making, while temporal gating allows for signal gating and, thereby, action selection (decision making) on a millisecond timescale. Moreover, precisely timed perturbations of the cortical network can systematically alter the behavior of the animal (Seidemann et al., 1998). However, more direct tests of this type of gating mechanism have not been performed thus far. As noted above, the temporal gating method exploits circuit level behavior, which makes this mechanism particularly interesting for *in vivo* experiments. Recent developments in optogenetic techniques, which allow selective activation/inactivation of specific groups of neurons (Zhang et al., 2007), could be used to test our proposal of temporal gating *in vivo* in a highly controlled manner.

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E62 延迟值落在在体测量范围内
作者报告｜把仿真中的 E/I 延迟与在体测量对齐，是机制可行性的一条支持性论据，但只覆盖延迟取值，不覆盖调节机制。

E63 作者的未来工作：短时程可塑性
作者报告｜作者提出的替代实现路径是用短时程可塑性突触动态调制 E/I 平衡，属于其自陈的未来工作而非本文已验证内容。

E60 作者划定的第一条边界
作者报告｜时序门控只在瞬变接近分界线时有效；靠近稳定不动点或过于接近背景涨落时可能失效，这限制了它的适用范围。

E61 边界的另一半
作者报告｜与前一条互为补充，说明机制的有效区间有明确边界，不能被当作普适门控。

E64 缺少直接检验
作者报告｜作者承认尚无直接检验，因此本文证据应限定为计算层面的机制可行性，不能写成已在体内得到验证。

E65 作者建议的体内检验路径
作者报告｜作者提出用光遗传技术选择性激活或失活特定神经元群来做体内检验，指明了后续实验的具体做法。

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